

Brad Berryman
Site Vice President

Susquehanna Nuclear, LLC
769 Salem Boulevard
Berwick, PA 18603
Tel. 570.542.2904 Fax 570.542.1504
Brad.Berryman@TalenEnergy.com



SEP 06 2017

U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

10 CFR 50.73

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388(387)/2017-006-00
UNIT 1 LICENSE NO. NPF-14
UNIT 2 LICENSE NO. NPF-22
PLA-7633

Docket No. 50-387
50-388

Attached is Licensee Event Report (LER) 50-388(387)/2017-006-00. The LER reports an event involving the loss of Secondary Containment due to a trip of a Zone II Exhaust Fan. This event was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented fulfillment of a safety function.

Should you have any questions regarding this submittal, please contact Mr. Jason Jennings, Manager – Nuclear Regulatory Affairs at (570) 542-3155.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.

A handwritten signature in black ink, appearing to be "B. Berryman".

B. Berryman (Plant Manager)

B. Berryman

Attachment: LER 50-388(387)/2017-006-00

Copy: NRC Region I
Ms. T. E. Hood, NRC Project Manager
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Mr. M. Shields, PA DEP/BRP

NRC FORM 366 (04-2017)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 03/31/2020	
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)								
1. FACILITY NAME Susquehanna Steam Electric Station Unit 2					2. DOCKET NUMBER 05000388		3. PAGE 1 OF 3	
4. TITLE Secondary Containment Declared Inoperable Due to Trip of Zone II Exhaust Fan								
5. EVENT DATE			6. LER NUMBER		7. REPORT DATE		8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR
07	10	2017	2017	- 006	- 00	09	06	2017
						FACILITY NAME Susquehanna Steam Electric Station Unit 1		DOCKET NUMBER 05000387
						FACILITY NAME		DOCKET NUMBER 05000
9. OPERATING MODE 1			11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)					
			☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)	
			☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)	
			☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)	
			☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)	
			☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)	
			☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)	
			☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☒ 50.73(a)(2)(v)(C)	
			☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(D)	
			☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)	
					☐ 50.73(a)(2)(i)(C)		☐ OTHER Specify in Abstract below or in NRC Form 366A	
10. POWER LEVEL 100								
12. LICENSEE CONTACT FOR THIS LER								
LICENSEE CONTACT C. E. Manges, Jr., Senior Engineer - Nuclear Regulatory Affairs							TELEPHONE NUMBER (Include Area Code) (570) 542-3089	
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT								
CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	REPORTABLE TO EPIX
B	VA	FSV	A499	Y				
14. SUPPLEMENTAL REPORT EXPECTED					15. EXPECTED SUBMISSION DATE			
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO					MONTH DAY YEAR 			
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)								
On July 10, 2017, at approximately 0250 hours, Reactor Building Zone II differential pressure was not maintained above the Technical Specification (TS) limit of 0.25 inch of vacuum water gauge (WG) due to Discharge Damper PDD27578B failing closed resulting in loss of the associated Reactor Building Zone II Exhaust Fan 2V205B. As a result, TS 3.6.4.1 Surveillance Requirement (SR) 3.6.4.1.1 was not met due to the loss of differential pressure. Reactor Building Zone II Exhaust Fan 2V205A was manually started and Reactor Building Zone II differential pressure was restored to greater than 0.25 inches WG by approximately 0301 hours. The condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(C) as an event or condition that could have prevented fulfillment of a safety function. The cause of the event was the failure of the ASCO solenoid valve. Based on further review, the ASCO solenoid valve has been determined to be less than adequate for its application in Reactor Building HVAC. There were no actual consequences to the health and safety of the public as a result of this event.								

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 2		05000388	YEAR	SEQUENTIAL NUMBER	REV NO.
			2017	- 006	- 00

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On July 10, 2017, at approximately 0250 hours, Reactor Building [EIS System Code: NG] Zone II differential pressure was not maintained above the Technical Specification (TS) limit of 0.25 inch of vacuum water gauge (WG) due to Discharge Damper PDD27578B [EIS System/Component Identifier: VA/DMP] failing closed resulting in loss of the associated Reactor Building Zone II Exhaust Fan 2V205B [EIS System/Component Identifier: VA/FAN]. Reactor Building Zone II Exhaust Fan 2V205A was manually started and Reactor Building Zone II differential pressure was restored to greater than 0.25 inches WG by approximately 0301 hours.

On July 10, 2017, at 0735 hours, this condition was reported (ENS #52844) in accordance with 10 CFR 50.72(b)(3)(v)(C) as an event or condition that, at the time of discovery, could have prevented the fulfillment of the safety function to mitigate the consequences of an accident by controlling the release of radioactive material. SSES has no redundant Secondary Containment System.

CAUSE OF EVENT

The cause of the event was the failure of an ASCO solenoid valve [EIS Component Identifier: FSV] associated with Discharge Damper PDD27578B. Based on further review, the ASCO solenoid valve has been determined to be less than adequate for its application in Reactor Building Heating Ventilation and Air Conditioning (HVAC) system.

ANALYSIS/SAFETY SIGNIFICANCE

Based on engineering analysis of the event, secondary containment could have performed its safety function of isolating as assumed in the accident analysis and of re-establishing 0.25 inches of vacuum WG (drawdown) within the assumed accident analysis time (10 minutes).

This event will not be counted as a safety system functional failure (SSFF) for the NRC performance indicator based on the engineering analysis supporting the system's ability to fulfill the safety function.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 2	05000388	YEAR	SEQUENTIAL NUMBER	REV NO.
		2017	- 006	- 00

CORRECTIVE ACTIONS

Key corrective actions include replacement of the failed solenoid valve with a more robust model.

COMPONENT FAILURE INFORMATION

Manufacturer: ASCO Services
Type: 3-way electrical solenoid valve
Process Fluid: Air
Model: ASCO 8320 Solenoid Valve, General Use Model 8320G003
Voltage: 120 VAC

PREVIOUS SIMILAR EVENTS

The following are recent LERs involving loss of secondary containment due to component failures:

LER 50-388(387)/2017-005-00, "Secondary Containment Declared Inoperable Due to Trip of Zone III Filtered Exhaust Fan," dated August 18, 2017.

LER 50-387(388)/2016-012-00, "Unit 2 HVAC Unable to Maintain Differential Pressure," dated May 26, 2016.

LER 50-387(388)/2016-003-00, "Unit 2 Zone 3 HVAC Unable to Maintain Differential Pressure," dated May 6, 2016.

LER 50-387(388)/2015-013-00, "Loss of Secondary Containment Due to Failure of Fans," dated February 2, 2016.

LER 50-387(388)/2015-012-00, "Loss of Secondary Containment Differential Pressure Due to Icing of the Intake Supply Plenum Screens," dated February 2, 2016.

LER 50-387(388)/2015-010-00, "Loss of Zone 2 During Unit 1 Reactor SCRAM," dated January 8, 2016.

LER 50-387(388)/2015-005-00, "Loss of Secondary Containment Due to Unit 2 Damper Alignment," dated June 25, 2015.

LER 50-387(388)/2015-002-00, "Loss of Secondary Containment due to failure of running fans," dated June 10, 2015.